Data Analysis & Visualization Report

For

U.S. Domestic Flights Performance Analysis (2015)

Version 1.0

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Track: Data Analytics

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1 Executive Summary

- This project analyzes over 5.8 million U.S. domestic flights from 2015 to identify key drivers of delays and cancellations. An interactive Power BI dashboard was developed to visualize performance metrics for airlines, airports, and routes.

- Key findings indicate that evening hours and Thursdays experience the highest disruption rates. While major hubs like ATL and ORD handle the most traffic, they also account for a high volume of cancellations. The analysis provides a strategic tool for stakeholders to improve operational efficiency, enhance resource management, and enable data-driven decision-making.

2 Project Planning

2.1 Objective

To analyze and visualize 2015 U.S. domestic flight data to uncover patterns in delays and cancellations, providing actionable insights for aviation industry stakeholders.

2.2 Scope

- **In-Scope:** Analysis of 2015 U.S. domestic flights, focusing on airline, airport, route, and temporal performance. The deliverable is an interactive Power BI dashboard and this report.
- **Out-of-Scope:** International flights, ticket pricing, and economic impact analysis.

2.3 Methodology

The project followed a structured data analysis lifecycle:

1. **Data Gathering:** Aggregated flight, airport, and airline datasets.
2. **Data Preparation:** Performed data cleaning and transformation using Power Query.
3. **Data Modeling:** Established a star schema to link the datasets.
4. **Visualization:** Developed a multi-page dashboard in Power BI.
5. **Insight Generation:** Extracted and documented key analytical findings.

3 Data & Modeling

3.1 Data Sources

The analysis is based on three primary CSV files:

- flights.csv: Transactional data for each flight (5.8M+ records).
- airports.csv: Dimension table with airport details.
- airlines.csv: Dimension table with airline names.

3.2 Data Preparation

Key preparation steps included handling missing values in delay columns by imputing them with zero, as a missing value indicates no recorded delay. We also ensured correct data types for numerical and date fields and filtered out irrelevant data to optimize performance. Furthermore, key DAX measures such as Cancellation Rate and On-Time Performance were created to facilitate the analytical insights.

3.3 Data Model

A star schema was implemented in Power BI.

- **Fact Table:** flights
- **Dimension Tables:** airports and airlines
- **Relationships:**
  - Flights [AIRLINE] -> airlines [IATA_CODE] (Many-to-One)
  - Flights [ORIGIN_AIRPORT] -> airports [IATA_CODE] (Many-to-One)
  - Flights [DESTINATION_AIRPORT] -> airports [IATA_CODE] (Many-to-One, Inactive)

4 Stakeholder Analysis

4.1 Key Stakeholders

- **Airlines:** Operations managers seeking to improve on-time performance.
- **Airports:** Management teams focused on resource allocation and operational efficiency.
- **Passengers:** Travelers aiming to make informed decisions about their journeys.
- **Aviation Regulators:** Government bodies monitoring industry performance.

4.2 Stakeholder Value Proposition

This analysis provides distinct value to each stakeholder group by answering their specific business questions:

Stakeholder	Value Proposition (Business Questions Answered)
Airlines	Which of our routes are underperforming? How do we compare to competitors on key delay metrics?
Airports	What are our peak hours for delays? Which airlines cause the most delays at our facility?
Passengers	Which airline should I choose for a specific route? What time of day is best to travel to avoid delays?
Regulators	What are the systemic causes of delays across the national air system (e.g., weather, security)?

5 UI/UX Design

5.1 Design Strategy

The dashboard was designed with a user-centric approach, focusing on clarity, simplicity, and intuitive navigation. The color scheme was chosen strategically: red and orange tones highlight negative metrics like delays and cancellations, while neutral colors are used for context. This "data storytelling" approach allows users to grasp key takeaways at a glance.

5.2 Dashboard Structure

The dashboard is a multi-page report, with each page dedicated to a specific analytical theme.

- **Home Page:** Serves as a navigation hub, providing a high-level overview and buttons to access detailed sections.
- **Analysis Pages:** Dedicated pages for Airline Performance, Airport Performance, Route Analysis, and Time Factors allow for a deep dive into each area.
- **Interactivity:** Slicers and cross-filtering are enabled across all pages, allowing users to dynamically explore the data (e.g., filter the entire report for a specific airline or airport).

6 Key Insights & Analysis

6.1 Airline Performance

- **On-Time Performance:** A significant variance exists among airlines. Legacy carriers like **Delta (DL)** and **Alaska Airlines (AS)** show better on-time performance, while low-cost carriers like **Spirit (NK)** and **Frontier (F9)** exhibit higher average delays.
- **Cancellations: American Eagle (MQ)** has the highest cancellation *rate*, indicating potential operational or fleet-specific issues.

6.2 Airport Performance

- **High Traffic & Cancellations:** Major hubs like **Chicago O'Hare (ORD)**, **Dallas/Fort Worth (DFW)**, and **Hartsfield-Jackson Atlanta (ATL)** record the highest number of total cancellations, largely due to their high volume of flights.
- **Average Delays:** Smaller, regional airports sometimes show high *average* delay times, suggesting they are more susceptible to network disruptions originating from larger hubs.

6.3 Route & Temporal Analysis

- **Busiest Routes:** Short-haul routes between major hubs, such as **San Francisco (SFO) to Los Angeles (LAX)**, are the busiest by flight volume.
- **Temporal Patterns:**
  - **Time of Day:** Delays accumulate throughout the day, peaking in the **evening (6 PM - 9 PM)**.
  - **Day of Week: Thursdays and Fridays** experience the highest volume of delays, likely due to the start of weekend travel.
  - **Seasonality:** Winter months (December, January, February) show a spike in weather-related delays and cancellations.

7 Conclusion & Recommendations

Conclusion

The analysis successfully identified critical patterns in U.S. flight delays and cancellations for 2015. The interactive dashboard provides a powerful tool for exploring these patterns. The primary drivers of inefficiency are a combination of airline-specific operations, airport congestion, and predictable time-based demand spikes.

Recommendations

For Airlines: Focus on improving operational resilience for evening flights. Airlines with high delay rates should review their turnaround schedules and contingency plans for high-traffic routes.

For Airports: Enhance resource allocation during peak evening hours and on Thursdays/Fridays. For airports with high weather-related delays, investing in advanced de-icing and runway management technology is crucial.

For Passengers: To minimize delay risk, travelers should consider booking morning flights and avoiding travel on Thursdays if possible.

8 Appendix

Submission Details

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Project Submitted To: Digital Egypt Pioneers Initiative (DEPI)

Tools & Technologies

- Data Transformation &Visualization: Microsoft Power BI (Version: 2.149.1203.0 64-bit (November 2025))
- Data Source: CSV Files (The flight, airline, and airport datasets were obtained from Kaggle. Dataset: U.S. Domestic Flights — 2015)